

SESSION 7: MODEL CARD + DATA SHEET

Transparency, Responsible Use, and Documentation of a Construction Digital Twin Dataset

This document presents a Model Card and an expanded Data Sheet for the Digital Twin Cognitive Learning Model. The objective is to make the model and, especially, its underlying dataset understandable, traceable, and responsibly usable by researchers, project managers, engineers, and future auditors.

1. What Are Model Cards and Data Sheets?

1.1 Model Card: the model's technical label

A Model Card is a standardized document that describes a machine learning model, its intended applications, evaluation results, limitations, risks, and ethical considerations. It provides a concise but structured way to communicate what a model can and cannot reliably do.

The concept was introduced by Margaret Mitchell and collaborators in 2018 to improve transparency and accountability in machine learning. A Model Card can be compared with a product label: it does not replace technical documentation, but it allows users to understand the most important characteristics before using a model.

1.2 Data Sheet: the dataset's provenance record

A Data Sheet documents the dataset behind a machine learning system. It should explain why the dataset was created, how the data were collected, what populations or projects are represented, how variables were defined, what preprocessing was performed, what information was excluded, what biases or limitations may exist, and which uses are appropriate or inappropriate.

For this project, the Data Sheet is particularly important because construction data are heterogeneous and may contain confidential commercial information. Project location, contract value, delays, change orders, soil conditions, and final costs can also introduce geographic, organizational, temporal, and project-type biases. Therefore, dataset documentation is not merely administrative; it is part of the scientific validity of the model.

2. Model Card — Digital Twin Cognitive Learning Model

2.1 Model Details

Field	Value
Model name	Digital Twin Cognitive Learning Model
Version	v1.0
Model type	Hybrid regression model: XGBoost + CatBoost + Case-Based Reasoning (CBR)
Owner / researcher	Jhon Wilbert Quiroz Quilla
Development context	Construction project cost and schedule optimization
Primary task	Prediction of project cost overruns and schedule delays
Model status	Research / prototype — pending full real-project validation

2.2 Intended Use

Field	Description
Primary purpose	Predict potential cost overruns and schedule delays in construction projects.
Target users	Project managers, construction managers, resident engineers, planners, contractors, and researchers.
Appropriate uses	Budget planning, early risk identification, scenario analysis, project monitoring, and decision support.
Decision role	Decision-support tool; it should complement professional engineering and project-management judgment.
Not intended for	Clinical decisions, structural safety certification, legal determinations, automatic contract decisions, or decisions where model uncertainty is ignored.

2.3 Training Data

Field	Description
Source	Technical project files, progress valuations, final accounts/liquidations, and related project records.
Estimated size	Approximately 40–100 projects; final count must be confirmed before publication.
Geographic scope	Lima Metropolitan Area, according to the current project definition.
Temporal scope	2015–2025.
Key variables	Planned cost, actual cost, planned duration, actual duration, change orders, location, project characteristics, and soil-related information where available.
Target variables	Cost overrun and schedule delay, preferably represented both as absolute values and normalized ratios.
Data sensitivity	Potentially confidential commercial and project information.

2.4 Evaluation Data

Field	Description
Split	70% training / 15% validation / 15% test.
Principle	The test set must remain isolated from model development and final tuning.
Preprocessing	Data splitting should occur before fitting imputation, scaling, feature selection, or other transformations.
Leakage prevention	Statistics learned from training data must not be calculated using validation or test observations.
Recommended enhancement	For a small dataset, report repeated cross-validation or bootstrap confidence intervals in addition to the final hold-out test.

2.5 Performance Metrics

Metric	Target / Interpretation
MAPE	Target < 12%; interpret cautiously when actual values are close to zero.

R^2	Target > 0.75; indicates the proportion of variance explained, not prediction accuracy by itself.
MAE	Report in meaningful project units, such as monetary value or days, whenever possible.
RMSE	Recommended to penalize large prediction errors more strongly.
Prediction interval / uncertainty	Recommended for communicating the range in which predictions may plausibly fall.

2.6 Ethical, Privacy, and Risk Considerations

Risk	Mitigation
Confidential project information	Anonymize identifiers, remove unnecessary commercial details, restrict access, and maintain controlled data storage.
Geographic bias	Report geographic coverage and evaluate whether performance changes across locations.
Project-type bias	Document which construction sectors and project sizes are represented and evaluate performance by project category.
Historical bias	Recognize that past projects may reflect outdated practices, prices, regulations, or management conditions.
Data leakage	Audit features to ensure that information unavailable at prediction time is not used.
Automation bias	Present the system as decision support rather than an autonomous decision maker.
User overconfidence	Display uncertainty, limitations, and validation status alongside predictions.

3. Data Sheet — Construction Project Dataset

The Data Sheet is the central focus of this session because model quality is inseparable from dataset quality. A high-performing algorithm trained on incomplete, biased, leaked, or poorly documented data can generate apparently accurate predictions that fail when deployed in a different project context. The following specification therefore extends the original dataset description into a more rigorous research artifact.

3.1 Motivation

Field	Description
Purpose	Train, validate, and evaluate predictive models for construction project cost and schedule management.
Research objective	Identify patterns associated with cost overruns and delays and support early risk assessment.
Creator	Jhon Wilbert Quiroz Quilla, based on construction project records.
Expected benefit	Improve planning, risk identification, forecasting, and evidence-based project management.
Unit of analysis	Construction project, with the possibility of extending the dataset to project phases or periodic observations in future versions.

3.2 Dataset Composition

Field	Description
Number of projects	Approximately 40–100; to be confirmed and frozen in the dataset release.
Geographic coverage	Lima Metropolitan Area, 2015–2025.
Project coverage	Construction projects represented in the available technical and financial records.
Core variables	Planned cost, actual cost, planned duration, actual duration, change orders, location, project characteristics, and soil-related variables where available.
Derived variables	Cost overrun ratio, schedule delay ratio, change-order density, project complexity indicators, and other features justified by domain knowledge.
Data structure	Structured tabular dataset, with project-level observations and documented variable definitions.
Missingness	Must be quantified per variable and per project before modeling.
Sensitive information	Company names, contract identifiers, personal information, and commercially sensitive fields should be anonymized or removed.

3.3 Variable and Target Definitions

Variable / Concept	Definition / Recommended treatment
Planned cost	Approved or baseline project cost defined before execution; document the exact source and price basis.
Actual cost	Final recorded project cost according to the available financial documentation.
Cost overrun	Preferably calculated as $(\text{Actual Cost} - \text{Planned Cost}) / \text{Planned Cost} \times 100$.
Planned duration	Baseline project duration in days or another clearly defined unit.
Actual duration	Observed duration from the selected project milestones.
Schedule delay	Preferably calculated as $(\text{Actual Duration} - \text{Planned Duration}) / \text{Planned Duration} \times 100$.
Change orders	Number and/or monetary value of approved changes; define the cut-off date used for prediction.
Location	Geographic information represented at an appropriate aggregation level to protect confidentiality.
Soil type	Geotechnical classification where reliably documented; missing or inconsistent classifications must be flagged.

3.4 Data Collection Process

Field	Description
Collection method	Systematic extraction from technical files, progress valuations, final accounts/liquidations, and related project documentation.
Collection period	2015–2025.

Selection criteria	Projects must contain sufficient information to define the target variables and the minimum required predictors.
Documentation	Each project should receive an anonymous project ID and a provenance record indicating the source documents used.
Quality control	Cross-check costs and durations between source documents; flag contradictory or incomplete records.
Confidentiality	Use agreements or authorized access procedures established with participating organizations.
Data lineage	Maintain a record of raw source, cleaned version, transformed version, and modeling dataset.

3.5 Inclusion and Exclusion Criteria

This section is essential for preventing hidden selection bias. Each excluded project should have a documented reason.

Criterion	Recommended rule
Include	Project has reliable planned and actual cost information.
Include	Project has reliable planned and actual duration or sufficient milestones to derive them.
Include	Key predictor fields meet the minimum data-quality threshold.
Exclude	Critical target information is missing or cannot be reconstructed reliably.
Exclude	Records contain unresolved contradictions that could materially alter the target.
Exclude	Duplicate project records without a defensible method for consolidation.
Review separately	Extreme projects or unusual project types should not automatically be removed; investigate whether they represent valid real-world variation.

3.6 Preprocessing and Feature Engineering

Stage	Method / Control
Cleaning	Remove duplicates, standardize units, correct formatting errors, and document every transformation.
Missing values	Quantify missingness; use imputation only after the train/validation/test strategy is established.
Outliers	Investigate domain validity before removing or winsorizing extreme observations.
Numerical variables	Standardization may be applied where required by the selected algorithm; tree-based models generally do not require it.
Categorical variables	Encode categories consistently and prevent unseen-category failures at inference.
Feature engineering	Create domain-relevant ratios such as change-order density and project complexity indicators.
Leakage control	Only use information that would have been available

	at the moment the prediction is intended to be made.
Versioning	Store dataset and transformation versions using a reproducible data-versioning mechanism.

3.7 Data Quality and Bias Assessment

Before model training, the dataset should undergo a formal quality audit. The audit should report the number of missing values, duplicate records, inconsistent units, extreme observations, category imbalance, temporal gaps, and project-type imbalance. It should also compare the distribution of key variables between training, validation, and test sets.

Potential issue	Audit question	Recommended evidence
Missing data	Which variables have the highest missingness?	Missing-value percentage by variable and project.
Geographic bias	Does the dataset represent only a narrow geographic context?	Performance and sample distribution by location.
Temporal drift	Have cost structures or construction practices changed from 2015 to 2025?	Performance by year or time period.
Project-size bias	Are large or small projects underrepresented?	Project-size distribution and stratified evaluation.
Target imbalance	Are severe overruns/delays rare?	Distribution plots and class/range statistics.
Data leakage	Could any predictor contain information generated after the prediction point?	Feature availability audit.
Sampling bias	Why were these projects available for inclusion?	Documented selection and exclusion criteria.

3.8 Appropriate and Inappropriate Uses

Appropriate use	Inappropriate use
Early-warning risk assessment	Automatic approval or rejection of projects
Budget and schedule scenario analysis	Structural safety certification
Project monitoring and prioritization	Clinical or medical decisions
Research on construction performance	Legal or contractual decisions without independent review
Decision support for project managers	Treating predictions as guaranteed outcomes

3.9 Dataset Limitations

- The estimated sample size of 40–100 projects is relatively small for a complex hybrid machine learning system and must be confirmed before final model evaluation.
- The current geographic scope may limit generalization to other regions, countries, climates, regulatory environments, or construction markets.
- Historical data from 2015–2025 may contain temporal drift in prices, regulations, technologies, procurement practices, and construction productivity.
- Confidentiality restrictions may prevent public release of the raw dataset, reducing external reproducibility.
- The quality of derived variables depends on the consistency and completeness of the original project records.

- Small samples increase the risk of unstable performance estimates; confidence intervals and repeated validation are therefore recommended.
- The model should not be assumed to generalize to project types that are poorly represented in the dataset.

4. Recommended Dataset Release Structure

To strengthen reproducibility while protecting confidential information, the project should maintain separate layers of data:

Layer	Content	Access
Raw data	Original project records and source documents.	Restricted / authorized personnel only.
Curated data	Cleaned and standardized records with anonymous IDs.	Restricted research access.
Modeling dataset	Final features and targets used for experiments.	Controlled research access; publish where legally possible.
Metadata / Data Sheet	Variable definitions, provenance, inclusion/exclusion rules, quality metrics, and version information.	Public or broadly shareable.
Synthetic / anonymized sample	Non-sensitive demonstration dataset for pipeline testing.	Public, if legally and technically appropriate.

5. Final Assessment

The Model Card establishes how the Digital Twin Cognitive Learning Model should be interpreted and used, while the Data Sheet establishes how the dataset should be understood, governed, and evaluated. The dataset is the foundation of the entire predictive system: if its provenance, variables, exclusions, missing values, temporal coverage, and biases are not documented, model metrics alone cannot establish scientific reliability.

The most important improvement is to move from an estimated description of the dataset to a verified dataset inventory. Before claiming final performance, the project should freeze the exact number of projects, define every variable, quantify missingness, document exclusions, verify that no future information leaks into predictors, and evaluate performance across relevant project and geographic subgroups.

In this way, the Data Sheet becomes more than a descriptive document: it becomes a governance mechanism that connects data provenance, model validity, ethical use, reproducibility, and responsible deployment.

6. References

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